ATHARV C. JOSHI

AI & Machine Learning Engineer | Robotics | Computer Vision

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PROFESSIONAL SUMMARY

Motivated AI and Machine Learning Engineer with hands-on experience in autonomous robotics, computer vision, deep learning, and IoT systems. Proven ability to design and deploy end-to-end AI solutions from model development to production deployment. Consistent top performer in national-level robotics competitions (IIT Madras, IIT Bombay, COEP Pune). Proficient in Python, C++, ROS2, PyTorch, OpenCV, and REST API development. Passionate about building scalable, intelligent systems that solve real-world problems.

TECHNICAL SKILLS

Programming Languages: Python, C++, JavaScript, HTML, CSS
AI / Machine Learning: PyTorch, TensorFlow, OpenCV, Deep Learning, Computer Vision, NLP, Supervised Learning, Unsupervised Learning
Robotics & Automation: ROS2, Autonomous Navigation, Sensor Fusion, SLAM, Gazebo, RViz, Path Planning, Obstacle Detection
Backend & APIs: REST APIs, Google Gemini API, OpenAI GPT API, Streamlit, Serverless Architectures, GitHub Pages
IoT & Embedded Systems: Raspberry Pi, Microcontrollers, Sensor Interfacing, Real-Time Systems, Hardware Debugging
Development Tools: Git, GitHub, Linux, VS Code, Cloud Dashboards, Simulation Environments

EDUCATION

Bachelor of Technology – Artificial Intelligence & Machine Learning *Expected: May 2027*
D. Y. Patil Agriculture and Technical University, Kolhapur, Maharashtra
Relevant Coursework: Artificial Intelligence, Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems

Senior Secondary (Class XII) – Padmashree Dr. Vitthalrao Vikhe Patil Sainik School, Pravaranagar *63%*
Secondary (CBSE Class X) – Nehru Montessori Senior Secondary School, Burhanpur *75%*

PROJECTS

SehatSaathi – AI-Powered Health and Wellness Platform *Python | PyTorch | OpenCV | Streamlit | OpenAI GPT API*

- Built a multi-condition AI diagnostic platform detecting skin, eye, oral, bone X-ray, and lung diseases using custom PyTorch deep learning models with transfer learning.
- Engineered image preprocessing and region-of-interest detection pipelines with OpenCV, improving diagnostic accuracy and reducing false positives by over 20%.
- Integrated OpenAI GPT API to deliver a personalized AI health coach providing lifestyle guidance and medical insights based on user inputs.
- Developed automated workout and nutrition planners generating weekly plans tailored to individual fitness goals and dietary requirements.

Autonomous Surveillance Rover *Python | C++ | ROS2 | Raspberry Pi | Computer Vision | Gazebo | RViz*

- Designed and built a fully autonomous rover using the ROS2 navigation stack integrated with on-board sensors for real-time navigation and area surveillance.
- Implemented perception modules for obstacle detection, terrain analysis, and adaptive speed control using camera feeds and range sensor data.
- Developed end-to-end simulation workflows in Gazebo and RViz to validate and iterate on navigation algorithms prior to hardware deployment.
- Optimized ROS2 decision-making nodes for efficient execution on embedded Raspberry Pi hardware while sustaining real-time performance requirements.

College Information Assistant – Conversational AI Chatbot *Python | Google Gemini API | REST APIs | GitHub Pages*

- Built a multi-turn conversational AI chatbot using the Google Gemini API to answer student queries on admissions, fees, scholarships, and deadlines.
- Designed modular backend REST APIs for FAQs and announcements, enabling real-time content updates without modifying core application logic.
- Deployed on GitHub Pages using a serverless architecture, achieving high availability and minimal maintenance overhead.

IoT-Based Smart Monitoring Systems *IoT Sensors | Microcontrollers | Raspberry Pi | Cloud Dashboards*

- Developed IoT systems for water quality monitoring, smart irrigation control, and laser-based perimeter security using microcontrollers and Raspberry Pi.
- Integrated sensor arrays with cloud dashboards for real-time data monitoring, event logging, and automated alert notifications.
- Implemented rule-based automation for irrigation pumps and security triggers enabling autonomous response to sensor threshold breaches.

ACHIEVEMENTS & AWARDS

- 2nd Place – Rover Domain, MindSpark 2025 (COEP Pune): Secured 4th place overall and 2nd in Rover Domain at L&T Tech Xcelerate for developing an autonomous navigation rover.
- 13th Rank Nationwide – RoboRun, Shaastra 2026 (IIT Madras): Ranked 13th nationally as part of Team AutoNavX, building an autonomous maze-solving robot using IR sensors and real-time navigation algorithms.
- Top 5 Finalist – Autodev Hackathon, Techfest IIT Bombay: Selected among top 5 teams nationally in an autonomous systems engineering hackathon.
- Cyber Suraksha Warrior of the Year 2025 – Quick Heal Foundation: Recognized for outstanding contributions to cybersecurity awareness and digital safety initiatives.

CORE COMPETENCIES

Team Leadership | Problem Solving | Technical Communication | Analytical Thinking | Time Management | Collaboration | Research & Development | Adaptability

LANGUAGES

English (Professional Proficiency) | Hindi (Native) | Marathi (Native)